

chromatin usually appears as a diffuse mass within the nucleus, with some denser clumps, including in regions of centromeres and telomeres. The less compacted, more dispersed interphase chromatin is called **euchromatin** (“true chromatin”) to distinguish it from the more compacted, denser-appearing **heterochromatin** (see Figure 16.23). Recent research has clarified our understanding of chromatin structure: For both types of chromatin, the basic organizing unit is the 10-nm fiber—nucleosomes joined by linker DNA. In heterochromatin, this 10-nm fiber is folded and bent back on itself to a much greater degree than in euchromatin, accounting for its denser appearance. Because heterochromatin is so compacted, it is largely inaccessible to the proteins responsible for transcribing the genetic information, a crucial early step in gene expression. In contrast, the looser packing of euchromatin makes its DNA accessible to those proteins, and the genes present in euchromatin are available for transcription.

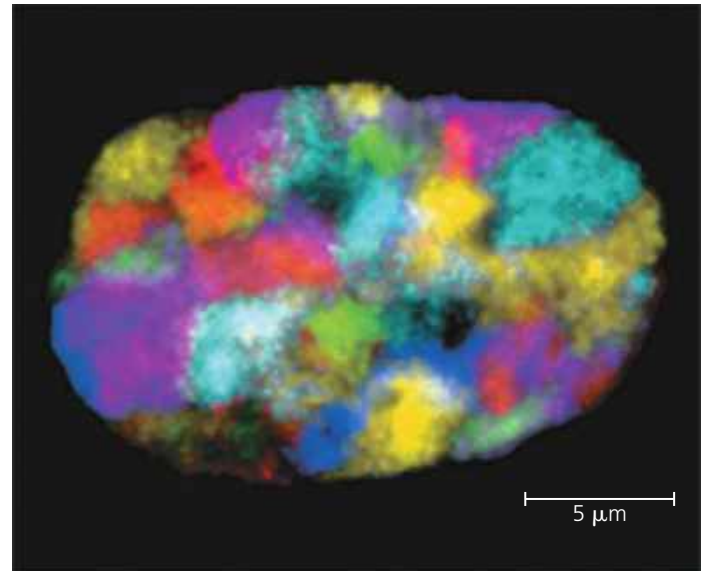
Early on, biologists assumed that interphase chromatin was a tangled mass in the nucleus, like a bowl of spaghetti, but this is far from the case. Although an interphase chromosome lacks an obvious scaffold, there are proteins that further organize the 10-nm fiber into larger compartments and smaller looped domains. Some of the looped domains appear to be attached to the nuclear lamina, on the inside of the nuclear envelope, and perhaps also to fibers of the nuclear matrix. These attachments may help organize regions of chromatin where genes are active. The chromatin of each chromosome occupies a specific restricted area within the interphase nucleus, and the chromatin fibers of different chromosomes do not appear to be entangled (**Figure 16.24a**).

As a cell prepares for mitosis, its chromatin becomes organized into loops and coils, eventually condensing into a characteristic number of short, thick metaphase chromosomes that are distinguishable from each other with the light microscope (**Figure 16.24b**).

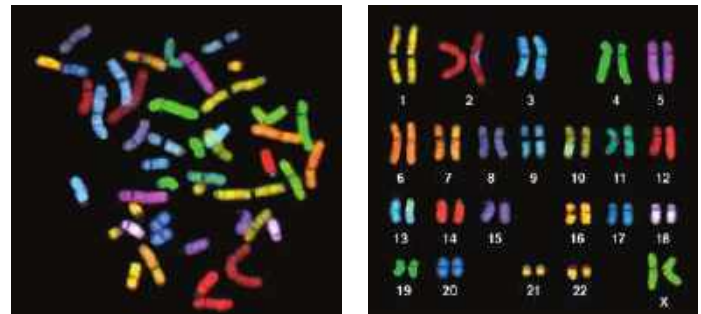
The chromosome is a dynamic structure that is condensed, loosened, modified, and remodeled as necessary for various cell processes, including DNA replication, mitosis, meiosis, and gene expression. Certain chemical modifications of histones affect the state of chromatin condensation and also have multiple effects on gene expression, as you’ll see in Concept 18.2.

In this chapter, you have learned how DNA molecules are arranged in chromosomes and how DNA replication provides the copies of genes that parents pass to offspring. However, it is not enough that genes be copied and transmitted; the information they carry must be used by the cell. In other words, genes must also be expressed. In the next chapter, we will examine how the cell expresses the genetic information encoded in DNA.

▼ **Figure 16.24 “Painting” chromosomes.** Researchers can treat (“paint”) human chromosomes with molecular tags that cause each chromosome pair to appear a different color.



(a) The ability to visually distinguish among chromosomes makes it possible to see how the chromosomes are arranged in the interphase nucleus. Each chromosome appears to occupy a specific territory during interphase. In general, the two homologs of a pair are not located together.



(b) These metaphase chromosomes have been “painted” so that the two homologs of a pair are the same color. On the left is a spread of treated chromosomes; on the right, they have been organized into a karyotype.

MAKE CONNECTIONS If you arrested a human cell in metaphase I of meiosis and applied this technique, what would you observe? How would this differ from what you would see in metaphase of mitosis? Review Figure 13.8 and Figure 12.7.

CONCEPT CHECK 16.3

1. Describe the structure of a nucleosome, the basic unit of DNA packing in eukaryotic cells.
2. How does euchromatin differ from heterochromatin in structure and function?
3. **MAKE CONNECTIONS** Interphase chromosomes appear to be attached to the nuclear lamina and perhaps also the nuclear matrix. Describe these two structures. See Figure 6.9 and the associated text.

For suggested answers, see Appendix A.